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Title: Did the 1918 pandemic cause a 1920 baby boom? Demographic evidence from neutral Europe

Running head: The 1918 pandemic and the 1920 baby boom

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Abstract:

In 1919–20, the European countries that were neutral in World War I saw a small baby bust followed by a small baby boom. The sparse literature on this topic has attributed the 1919 bust to individuals postponing conceptions during the peak of the 1918–20 influenza pandemic and the 1920 boom to them recuperating those conceptions. Using data from the six large neutral countries of Europe, we present novel evidence contradicting that narrative. In fact, the subnational populations and maternal birth cohorts whose fertility was initially hit hardest by the pandemic were still experiencing below-average fertility in 1920. Demographic evidence, economic evidence, and a review of post-pandemic fertility trends outside of Europe suggest that the 1920 baby boom in neutral Europe was caused by the end of World War I, not by the end of the pandemic.

Keywords: 1918 influenza pandemic; 1920 baby boom; historical demography; fertility; pandemics; mortality-fertility nexus; World War I

Introduction

The classical framework of how human populations respond to mortality crises suggests that fertility should first fall, then rebound in the wake of mass death (Livi Bacci 2000; Heuveline and Poch 2007; Nobles et al. 2015). This is expected to be particularly true in the case of mortality crises that disproportionately affect young adults, such as the 1918–20 influenza pandemic (see Langford 2002; Saglanmak et al. 2011; van Wijhe et al. 2018). Therefore, just as that pandemic was followed by the Roaring Twenties—a decade of cultural innovation and relative economic prosperity in much of the Western world—one might expect that it was followed by a baby boom.

The 1920s was a time of historically unprecedented low fertility in Europe (Van Bavel 2010), but the basic demographic data does support the narrative of a short-term, post-pandemic bust-boom cycle well. Figure 1 compares the relative crude birth rates in 1918, 1919, and 1920 (compared to 1915–17) in France, Hungary, Germany, and the seven European countries militarily unaffected by the war. The belligerent countries saw very high upticks in fertility in both 1919 and 1920—because the war suppressed their fertility so gravely (see e.g. Vandenbroucke 2014)—but with the exception of Spain, the neutral countries see a clear bust-boom pattern over those years. The conclusion in the recent literature has been that World War I played little role in the neutral countries' fertility trends in this period and that their 1920 baby booms are due to post-pandemic fertility recuperation (Mamelund 2004, 2012; Bloom-Feshbach et al. 2011).

The mechanisms by which the pandemic could have caused a baby boom are reasonable on face. In populations across the world, there was a marked birth deficit roughly nine months after the pandemic's peak (Mamelund 2004; Bloom-Feshbach et al. 2011; Chandra and Yu 2015a, 2015b; Chandra et al. 2018; Dahal et al. 2018;

Wagner et al. 2020; Kadt et al. 2021). This indicates that many conceptions were forgone during the pandemic, and one can assume that a sizable proportion of those conceptions were recuperated as the pandemic waned. There was also a surge in stillbirths around the time of the pandemic's peak (Reid 2005; Nishiura 2009; Chandra and Yu 2015b; Helgertz and Bengtsson 2019; Floris et al. in press; cf. Tripp et al. 2018; Khare et al. 2020), and this may have been associated with some post-pandemic live births. Child death is known to increase the subsequent probability of maternal conception through the counterfactually early cessation of lactational amenorrhea and the desire to "replace" the child lost (Palloni and Rafalimanana 1999; Lindstrom and Kiros 2007; Reher et al. 2017). Stillbirths may also trigger these two mechanisms. Additionally, there is some evidence that individuals can experience increased fertility in response to death in their communities independent of how that mortality may have dampened their fertility (Rutayisire et al. 2013; Nobles et al. 2015; Broussard and Weitzman 2020; Smith-Greenway et al. 2022).

However, the analysis of the 1918–20 pandemic and subsequent fertility trends has mostly been limited to time series studies. In only observing that the 1920 boom followed the 1919 bust, these analyses do not make a strong causal case that the bust caused the boom (see Ní Bhrolcháin and Dyson 2007). In fact, two of the most detailed studies of the pandemic's impacts on family formation and fertility—Underwood's (1984) study of Guam and Herring's (1993) study of a Cree and Métis community in Manitoba, Canada—frame the post-pandemic recovery of fertility as a process of moderately-paced adaptation, rather than as a one-year boom. Crucially, the existing literature does not test a plausible alternative hypothesis: that the pandemic caused the 1919 baby bust but some exogenous event, perhaps the end of World War I, caused the 1920 baby boom independently of the bust.

The lack of suitable microdata available at present means that these competing hypotheses cannot be tested at the individual level, but they can be assessed at the aggregate level by comparing the magnitude of the 1919 bust and 1920 boom across population subgroups (Mamelund 2004). If one assumes that the boom served as direct recuperation for the bust (the *recuperation hypothesis*), relative fertility in 1919 and 1920 should correlate negatively; geographic and cohort-specific subgroups that experienced a stronger bust in 1919 should have seen a stronger boom in 1920. On the other hand, the relative birth trends could correlate positively. Subgroups that experienced a stronger bust in 1919 may have also experienced a bust in 1920 because the 1920 boom had an exogenous source and the pandemic had a persistent effect whereby it dampened fertility for at least two years (the *persistent dampening hypothesis*). Mamelund (2004) reported evidence for the recuperation hypothesis based on geographic data from Norway, but to our knowledge, this method has not been employed in any other analysis of the pandemic.

In this article, we conclude that the persistent dampening hypothesis is a better interpretation of the available data from neutral Europe than the recuperation hypothesis is. Our analysis in support of this conclusion proceeds in four stages—each with a different degree of certainty—but we find that they collectively provide strong support for this new historical narrative about the pandemic. First, we test the recuperation and persistent dampening hypotheses using subnational fertility data from six countries and maternal cohort-specific fertility data from three countries. In doing so, we revisit Mamelund's (2004) work on Norway and extend it to five additional countries and an additional axis of population stratification. Second, we consider two alternate interpretations of these main results: that they are explained by the post-pandemic remarriage of widowers or by selective mortality during the pandemic. Third,

we review what non-European demographic data show about the fertility in the same period. Fourth, we use European economic data to bolster our suggestion that the 1920 baby boom in neutral Europe was caused by the economic effects of the end of World War I, rather than the end of the pandemic. Finally, we discuss the relationship of our findings to other areas of pandemic studies.

Data and methods

For our geographic analysis, we obtained annual subnational live birth counts for 1916–20 for Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain. These comprise all of the large European populations that were neutral during World War I, and our focus on neutral populations is based on the lack of war as an overridingly influential factor. Iceland was also neutral in the war, but subnational analysis for Iceland yields very small sample sizes. Iceland is also an unusual case because the pandemic was successfully quarantined to the west of the country until 1921 (Summers et al. 2013). Subnational birth count data was available at the municipal level for the Netherlands, at the province level for Sweden, and for the urban (or capital) and rural (or non-capital) parts of provinces for the four remaining countries. For our cohort analysis, we obtained annual Lexis fertility rates for 1916–20 in Sweden and Denmark and converted age-period fertility rates to Lexis rates for those years in Norway.

Geographic fertility anomalies for 1919 and 1920 were established by standardising the given annual birth count against the average annual birth count for 1916–18 in the same jurisdiction. Cohort fertility anomalies for 1919 and 1920 were established by subtracting the given period-cohort fertility rate (i.e. vertical Lexis parallelogram rates) from the average fertility in the same age-range Lexis

parallelograms in 1916–18. In our geographic analysis of the Netherlands, we analyse only those municipalities with 250 or more births per year at baseline, in order to exclude the erratic annual fertility swings experienced by the country's smallest municipalities. However, we conduct sensitivity analysis to check whether this exclusion criterion affects our findings. Data for 1918 is included in the birth baseline of the main analysis because the effect of the pandemic on live births seems to have been negligible in 1918, especially in comparison to 1919 (see Figure 1; Chandra et al. 2018), but 1918 is removed from the baseline in robustness checks. Additionally, we test whether adding stillbirths to our analysis affects our results.

At the time of the pandemic, vital registration completeness varied greatly within the countries of interest (see e.g. Ramiro Fariñas 1998), and we similarly expect subnational heterogeneity in the practices of recording a birth as live versus stillborn (see Davis 2009). However, these two sources of error should have remained fairly stable at the local level over 1916–20. Raw birth counts are used in the main geographic analyses instead of fertility rates because of denominator uncertainty. Even where annual subnational population estimates are, for example, reported in annual statistical yearbooks, the validity of these estimates is not always clear. Furthermore, interpolations between pre- and post-pandemic census counts are distorted by any intercensal fluctuations in rates, including due to the pandemic itself. However, we test whether our geographic findings are robust to changes in subnational exposures to fertility by using published annual population counts as provisional denominators in the Netherlands and exponential interpolations as provisional denominators in the other five populations.

All data and code utilised are provided as supplementary materials for this article, with the exception of the data underlying our economic analysis for Spain. That

data can be requested from the researchers who digitised it. Detailed information about our data sources (e.g. Brunborg and Mamelund 1994; Gómez-Tello et al. 2019; Boonstra 2021) and data handling are also provided in our supplementary materials.

Main results

Figure 2 compares the relative subnational birth counts observed in Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain in 1919 and 1920 compared to 1916–18. In each country besides Sweden, the anomalies correlate with each other strongly (adjusted R^2 = 0.338–0.664) and positively ($p < 0.001$, n = 35–112). This provides clear evidence in support of the persistent dampening hypothesis and clear evidence against the recuperation hypothesis. The regions that saw the strongest baby booms in 1920 are those that had seen minimal baby busts or fertility that was actually above-average in 1919. In Norway, Denmark, the Netherlands, and Switzerland, the regions with the strongest 1919 baby busts tended to see a baseline number of births in 1920, rather than a baby boom. In Spain, the regions with the strongest 1919 baby busts tended to also see baby busts in 1920—in fact, ones of a similar magnitude as in 1919. In Sweden, the available geographic data—which is at the highest level of aggregation of any of the six countries of interest—provides support for neither the persistent dampening hypothesis nor the recuperation hypothesis ($p = 0.674$, n = 25).

Supplementary analysis indicates that these results are robust to excluding 1918 from the birth count baselines (Figure S1) and to weighting each regression by the absolute size of the annual birth baseline (Figure S2). In the case of the Netherlands, the significance of our finding is not sensitive to our small-municipality exclusion criteria (Figure S3). Additionally, the finding that the relative fertility

anomalies in 1919 and 1920 are positively correlated holds if one uses estimates of local crude birth rates, instead of the recorded local birth counts (Figure S4). It also holds when adding stillbirths to the live birth counts in the countries with detailed stillbirth data available, namely Norway and Switzerland (Figure S5). We expect that this would also be the case in all other populations of interest, given that stillbirths tend to comprise a small proportion of all births, even in 1918–19.

Figure 3 compares the national female cohort-specific fertility rates (CSFR) observed in Norway, Sweden, and Denmark in 1919 and 1920 compared to 1916–18. In general, the 1920 baby boom in each country was concentrated among women who were aged 20 to 30 at the time, whereas the 1919 baby bust was concentrated among women who were aged 30 to 45. This is especially clear in Norway and Sweden. Women at reproductive ages under 25 (or 30 in Sweden) tended to see a net boom over the two years, whereas women at older reproductive ages tended to see a net bust. In other words, the cohorts that tended to see the strongest baby busts in 1919 tended to see the smallest baby booms in 1920. This finding is also robust to using 1916–17 (Figure S6) or 1915–17 as the fertility baseline (Figure S7). This provides support for the persistent dampening hypothesis in the same way as our geographic results.

Discussion

In all nine subgroup analyses—the panels of Figures 2 and 3—the evidence suggests that the 1920 baby boom was not recuperation for the 1919 baby bust. Most regions and younger reproductive cohorts experienced both a bust in 1919 and a boom in 1920, but populations with larger busts in 1919 did not see larger booms in 1920. In fact, the opposite seems to hold. In eight of nine analyses, the populations

whose fertility was hit hard in 1919 also tended to experience below-average fertility in 1920. Therefore, the pandemic seems to have dampened the fertility of some groups in these countries for at least two years after its peak.

Our geographic results for Norway directly contradict those of Mamelund (2004). The conclusion in that article in favour of the recuperation hypothesis seems to have unfortunately been drawn in error. In Table 8, Mamelund (2004) reports that the subnational birth rate anomalies in Norway in January to September 1919 correlate positively with those in October 1919 to September 1920. This is consistent with our findings in Figures 2 and S4 and supports the persistent dampening hypothesis. Mamelund (2004) interprets the sign of the correlation incorrectly—as though it were negative and therefore as support for the recuperation hypothesis.

Various mechanisms seem likely to have played some role in the persistent dampening we observe. Individuals whose spouses died in the pandemic may not have remarried until several years after the pandemic, if ever. Other individuals may have suffered pandemic sequelae (e.g. McCall et al. 2008; Strange 2022; cf. Dourmashkin et al. 2012) or gained new care responsibilities for family members that caused them to delay marriage or childbearing, perhaps permanently. The pandemic may have also induced new perceptions in individuals that discouraged them from marrying or having children, at least in the medium term. Such perceptions could include existential uncertainty (see Outka 2019) or economic insecurity (see Karlsson et al. 2014; Galletta and Giommoni 2022). Mamelund (2004) found that relative fertility in 1919 and 1920 still correlated strongly in Norway when controlling for excess mortality in the second half of 1918. This suggests that the pandemic dampened fertility largely through such perceptions, rather through than the impacts of its mortality. The fact that, in Sweden, Norway, and Denmark, 1919 fertility was

dampened the least among the younger cohorts that faced the highest mortality from the pandemic (Figure 3) also suggests an importance of such perceptions for the dampening observed.

Nevertheless, a 1920 baby boom did happen at the national level in five of the countries of study and in much of Spain, the sixth country in our analysis. This raises the question of what caused that boom. France, Germany, the United Kingdom, non-Māori New Zealand, and Hungary also saw fertility rise significantly above their long-term baseline trends in 1920 (see Van Bavel 2010; Vandenbroucke 2014; Wilson et al. 2019). This phenomenon is poorly studied—especially in comparison to the baby boom that followed World War II (cf. Van Bavel and Reher 2013; Sandström 2014; Gauvreau et al. 2018)—but it seems likely that the baby booms in those countries were recuperation following the severe impacts of World War I on those countries' lives, families, and economies.

In the absence of a competing hypothesis, we suggest that the 1920 baby boom in neutral Europe was also a reaction to the end of World War I. During the war, the fertility rates in those countries did not fall much below their long-term trends, if at all (see Mamelund 2004; Van Bavel 2010), but the war still had impacts on individuals' lives and livelihoods. Increases in food prices and decreases in real wages were common in each country during the war, and the economy rapidly improved in the immediate post-war period (De Jong 2005; Ahlund 2012; Straumann 2015; Pedersen 2017; Gómez-Tello et al. 2019; see also Butie et al. 2020), although there was then a global deflationary crisis in 1920–21 (see e.g. Montgomery 1955; Velde 2022). Therefore, it is plausible that the neutral countries of Europe experienced a baby boom in 1920 as the result of a sudden improvement in relative economic conditions due to the end of the war. It is also possible that a post-war mood of celebration boosted

fertility. In fact, it is thought that the celebrations of the November 1918 Armistice contributed to the pandemic caseloads in some locations (e.g. Crosby 1989, p. 85; Bulling 1991, pp. 16–7). However, quantitatively linking the high spirits from the end of the war in and of themselves to a baby boom is likely impossible. It suffices to say that the boom was due to the effects of the end of the war in general.

Alternative explanations

The role of widower remarriage

Previous work considering post-pandemic demographic trends has suggested that there was a large wave of pandemic widows and widowers remarrying in 1919 (Herring 1993; Muñoz-Pérez and Recaño-Valverde 2011). Additionally, two reviewers of this article suggested that the older widowers marrying younger women might explain the relatively high post-pandemic fertility of younger women (Figure 3). If this were the case, it would suggest a strong causal role of the pandemic in the 1920 boom, contrary to our overall argument. We suggest that widower remarriage (and nuptiality more generally) had a limited role in the 1920 baby boom in each of the six countries of study.

In the cases of Norway and Sweden, pre-existing literature on this topic supports our argument. In Norway, widows were constrained from remarrying for one year after the death of their husband by law and widowers usually did not remarry for one year after the death of their wife by custom (see Mamelund 2004). This suggests a limited role for the remarriage of widows and widowers in Norway's 1920 baby boom. The fact that the number of marriages of non-widowed individuals in 1919 were down 15% on 1917 suggests a limited role for nuptiality in the 1920 baby boom in Norway in general. In Sweden and Denmark, widow(er) remarriage customs were similar to

those in Norway (Gaunt and Löfgren 1981, pp. 56–7). Indeed, Swedish crude marriage rates fell modestly from 1917 to 1919 and did so to an extent that appears uncorrelated with mortality during the main wave of the pandemic (Boberg-Fazlic et al. 2021, Figure C2). There was a marriage boom in 1920 in both Sweden and Norway but there was a limited window for this to affect fertility in 1920 itself.

Another line of evidence comes from the fact that in Spain and Switzerland, excess nuptiality in 1919 does not correlate with excess fertility in 1920. Assuming that the communities with the most relatively low fertility in 1919 had the highest rates of pandemic widowhood, one can deduce from Figure 2 that the communities with the most pandemic widows and widowers actually had the smallest baby booms in 1920, but directly testing this issue where possible is also fruitful. Figure S8 shows that relative number of marriages in 1919 compared to 1917 does not correlate with the 1920 fertility anomaly across the rural and urban parts of Spanish provinces ($n = 98$). This is true whether looking at the 16% of marriages including at least one widow ($p = 0.586$) or at the 84% of marriages of two non-widows ($p = 0.723$). In Switzerland, marriage counts by canton are not further disaggregated by civil status, but the relative number of all marriages in 1919 compared to 1917 also does not correlate with relative fertility in 1920 across cantons ($p = 0.507$; $n = 25$; Figure S9).

In Spain, Switzerland, Denmark, and the Netherlands, comparing the numbers of excess marriages in 1919 and excess births in 1920 also suggests that post-pandemic marriages, especially by widowers, could have only caused a modest proportion of the excess births in 1920. In Spain, there were only 10,002 more marriages involving widowers in 1919 than 1917, compared to 21,200 more births in 1919 than 1917. Assuming that 20–40% of those excess marriages yielded a birth within the first year (see Dribe et al. 2017, Table 2) and that none of the women in

those marriages would have had children in 1920 otherwise (though an existing marriage, a different new marriage, or illegitimately), one can estimate that only 9–19% of the excess births in 1920 in Spain were due to post-pandemic widower remarriage. This is a ceiling on this estimate, due to our second assumption being conservative and the fact that marriages are more likely to be officially recorded than births.

Meanwhile, applying our two simple assumptions to Switzerland suggests that only 2–4% of the excess births in 1920 in that population were due to widower remarriage, based on only 970 excess marriages involved widowers in 1919 and 9,126 excess births in 1920. In the Netherlands, contemporary statistical yearbooks group the remarriage of widowers and male divorcees together. However, even if one assumes that there were only a negligible number of divorcees within that group, making the two assumptions above leads one to estimate that only 2–3% of the excess births in 1920 in the Netherlands were due to widower remarriage, based on 1,565 excess marriages involved widowers and male divorcees in 1919 and 19,875 excess births in 1920. In Denmark, the contemporary statistical yearbooks do not report marriage counts by civil status. However, even if one assumes that *all* 4,263 excess marriages in 1919 involved a widower, making two assumptions above leads one to estimate that widower remarriage can only explain 11–22% of the 7,919 excess births in 1920.

This review of post-pandemic marriage patterns in neutral Europe is far from complete. Yet, it seems fairly clear that the 1920 baby boom cannot be explained by a surge in the reproductive output of newly widowed men. In fact, the literature on Norway and Sweden and our rough calculations for Denmark suggest that the baby boom in neutral Europe may not have stemmed in large part from post-pandemic

nuptiality of any kind. If this is the case, it suggests that children born in 1920 in neutral Europe were disproportionately born to longer-married parents. In turn, this could suggest that children born in 1920 were disproportionately higher-order births—although the fact that the boom was concentrated among younger mothers (Figure 3) complicates this. However, this possibility is supported by the finding of Boberg-Fazlic et al. (2021) that in rural Sweden—where that country's baby boom was concentrated as a whole—births between September 1919 and September 1921 were disproportionately driven by higher-parity births rather than first-order births (see Table B5). This issue is returned to in our discussion of this work's relevance to other literature on the pandemic.

The role of selective mortality

It has also been suggested that selective mortality could explain our main result that relative fertility in 1919 and 1920 correlate positively. In general, the selectivity of crisis mortality has important consequences for post-crisis population dynamics (Watkins and Menken 1985; Thornton et al. 1991). In the case of the 1918–20 pandemic, one can imagine that the pandemic disproportionately killed individuals who would have been unlikely to have children in the post-pandemic period, thereby boosting post-pandemic fertility rates purely by diminishing the denominator.

It is unlikely that this alternative story explains our result for three reasons. First, the positive correlation between relative subnational busts and booms is present whether one examines fertility rates or raw birth counts (Figures 2 and S4). Second, even when looking at birth rates, the elevated fertility in 1920 did not generally extend in 1921–22 (see Figure S10). If the baby boom in 1920 was largely a result of denominator change, the boom would have persisted into later years in the absence

of another radical change in the denominator, all else being equal. Third, the selective aspects of the pandemic's mortality may have acted to decrease crude birth rates, rather than increase them. The pandemic's distinctive pattern was to disproportionately kill individuals at peak reproductive ages (see e.g. Saglanmak et al. 2011). Additionally, contrary to the previous narrative that the pandemic was "egalitarian" with respect to class, most quantitative analysis has found that the pandemic disproportionately killed people from poorer backgrounds (Klein 1973; Mamelund 2006; Grantz et al. 2016; Bengtsson et al. 2018; Wilson et al. 2018; see also Mamelund et al. 2021). Since the pandemic struck at a time when fertility tended to be higher among the poorer populations (i.e. unskilled labourers) in at least some of the countries of interest (Bras 2014; Dribe and Scalone 2014; Dribe et al. 2014; Jaeggi et al. 2022), its socioeconomic selectivity dimension likely further dampened fertility rates. Also, while it has been suggested that the pandemic disproportionately killed people infected with tuberculosis (Noymer and Garenne 2000; Noymer 2009; cf. Sawchuk 2009; van Doren and Sattenspiel 2021), it is not clear that people with tuberculosis tended to have lower fertility on average than people without it, especially given confounding of tuberculosis status with socioeconomic status.

Further results

Post-pandemic fertility beyond Europe

Our suggestion that the 1920 baby boom in neutral Europe was a result of the end of World War I is also supported by the near-complete absence of a 1920 baby boom outside of Europe. Figure 4 displays the change in the crude birth rate from 1915–17 to 1920 for a range of national populations throughout the world, including each of the non-belligerent European countries in Figure 1. The non-European

populations shown comprise all 12 populations with data available in Chesnais (1992), plus 5 selected populations with data available in *International Historical Statistics* (Palgrave Macmillan 2013), Cyprus (Verropoulou 1997), and Korea (with rates calculated from Kim 1966, p. 7, 21). Among the populations in neutral Europe, all but Spain had notably elevated fertility in 1920. Among the populations outside of Europe, all but Japan and Egypt had average or below average fertility in 1920.

This geographic disparity is made even clearer by the fact that the elevated value in Figure 4 for Egypt may only be an artefact of changing birth registration completeness. Between the 1907 and 1917 censuses in Egypt, the registered crude birth rate and the censual ratio of children under age 5 to women aged 15–49 fell to a similar degree, but between the 1917 and 1927 censuses, the children to women ratio stagnated while the registered birth rate soared, especially in cities and beginning in 1920 (see Kiser 1944; see also El-Badry 1955). Assuming that the 1927 census did not have significantly poorer coverage of children than the 1917 census, a clear interpretation of this divergence is that the persistent rise in the Egyptian birth rate beginning in 1920 is due to rapidly improving birth reporting, rather than an increase in fertility.

In fact, one might interpret the large number of non-European populations that had below-average fertility in 1920 as evidence for the persistent dampening hypothesis and its corollary that, in the absence of the effects of World War I, Europe would have also seen relatively low fertility in 1920 at the national level. Drawing this conclusion in general seems premature at the moment. Like in Western Europe, the post-war period saw secular, rapid fertility declines in the United States, Canada, Australia, and New Zealand (Caldwell 2006; Van Bavel 2010)—and it is not clear how much of the difference between fertility in 1915–17 and 1920 in those and other

populations is due to secular changes, rather than any dampening effects of the pandemic.

The fact that the directly-ruled parts of British India saw such unusually low fertility in 1920 (see also Mills 1986; Dyson 1989), as well as in 1921–22, may be more suggestive. Although the pandemic death toll in British India is greatly disputed (Davis 1951; Mills 1986; Hill 2011; Chandra et al. 2012; Tumbe 2020), the range of published estimates all suggest that it likely experienced the highest proportional death toll of any large national population in the world. Therefore, British India is exactly the population that one would expect to have clearly and strongly below-average fertility after the pandemic if the pandemic did have persistent dampening effects on fertility. This argument is tentative, as it is not understood what effect the pandemic had on vital registration completeness in British India (cf. Davis 1951, p. 237) nor how severely the Non-Cooperation Movement disrupted the quality of colonial statistics in 1921–22 (see Chandrasekhar 1972, p. 33). Indeed, the same concern can be levied about the effect of the Sam-il Movement for independence on vital statistics in Korea in 1919–20 (see Kim 1966, p. 8; Baldwin 1979, p. 149). However, it is further suggestive that the directly-ruled districts of British India with the highest influenza-attributed mortality rates in 1918–19 had significantly elevated female labour force participation rates in the service sector at the 1921 census (Fenske et al. 2022). Whether one interprets labour force participation as a cause or an effect of a slump in fertility, this result coheres with the suggestion that the pandemic dampened fertility in British India by a large amount and potentially for more than two years.

A few non-European populations seem to have clearly experienced abnormally high birth rates in 1920, as opposed to the case of Egypt. Taeuber (1958, pp. 232–3) reports that evidence of a 1920 baby boom in Japan shows up both in official vital

rates and in mid-century population pyramids, although she finds the boom ‘puzzling’. Mid-century censuses from South Africa also indicate a relatively large 1920 birth cohort (Kadt et al. 2021), and *International Historical Statistics* reports that the British Caribbean colonies of Barbados and Jamaica had crude birth rates in 1920 that were more than 20% higher than in 1915–17 (Palgrave Macmillan 2013). Mexico also experienced a large baby boom starting in 1920 (Zavala de Cosío 1992), but this came at the end of decade of revolution during which the pandemic contributed only a small proportion of the overall demographic disruption (McCaa 2003).

These cases warrant further investigation but may be attributable to the fact that all of these populations were militarily involved in World War I—either as independent powers or British colonies—or were experiencing other very exceptional circumstances. Additionally, Ohbuchi (1998, pp. 339–40) frames the crude birth rate in Japan in 1920 as simply the peak of a gradual increase in that indicator that began around 1890 (see also Chandra and Yu 2015b). The 1918–20 pandemic was a global phenomenon, with almost no national exceptions (cf. McLane 2013). Indeed, the pandemic’s havoc on each of the non-European populations in Figure 4 is at least partially documented (e.g. McQueen 1976; Langford and Storey 1992; Killingray 1994, 2003; Verropoulou 1997; Lee et al. 2007; Gealogo 2009; Hsieh 2009; Lim 2011; Maureira 2012; Humphries 2013; Hayami 2015; Botey Sobrado 2017; Summers et al. 2018; Cristina et al. 2019; Carbonetti and Rivero 2020; Tumbe 2020; Rose 2021). Any pandemic-induced baby boom should be visible on a similarly global scale as the pandemic itself. Since the 1920 baby boom seems to have occurred almost only in Europe, we suggest that it must have a non-pandemic, Europe-specific explanation: World War I.

Testing World War I's effect on the baby boom

Our evidence that the 1920 baby boom was not caused by the pandemic serves as indirect evidence that the end of World War I caused it instead. Our argument here is a tentative one that since the pandemic was not the culprit, the war must have been, as the only other likely suspect. Such arguments by exclusion can be correct, and they have played an important role in the field of historical demography (e.g. McKeown and Record 1962). However, in our case, some economic data from the end of the war and immediate post-war period can also be brought to bear on the new historical narrative we propose.

We hypothesise that communities whose economies recovered the fastest in the year after the end of World War I are those which had the largest 1920 baby boom, controlling for the local 1919 baby bust. Unfortunately, the post-war economic recovery is difficult to measure at the local level. Price and wage data seems to have rarely been collected systematically. They also seem to have been rarely reported together, which is a problem because prices and wages jointly affect household purchasing power. Increasing wages could be a genuine sign of prosperity or simply a (partial) response to inflation. However, in the absence of simultaneous data on both variables, we assume that relative price increases are a negative economic indicator and relative wage increases are a positive economic indicator. Therefore, we interpret a negative association between price increases in 1919 and relative fertility in 1920 as evidence that the prevailing economic fluctuation was a major determinant of fertility in 1920.

Spain presents a rare case in which price data for household goods was collected continuously and at the local level in this period. In particular, biannual price data is available for a range of 21 household goods (plus the cost to rent a single room) at the same geographic level as fertility data, i.e. the rural and urban parts of

provinces (Gómez-Tello et al. 2019). Therefore, we have tested whether price changes between the winter of 1918–19 and the winter of 1919–20 correlate with relative fertility in 1920, controlling for relative fertility in 1919. Additionally, we have tested whether there is a statistically significant interaction between price change and 1919 fertility in their association with 1920 fertility. One would theoretically expect fertility to be more responsive to price changes in communities in which the pandemic had not dampened fertility strongly.

Figure 5 shows the most supportive results from this analysis. We find a clear interaction effect whereby increases in the price of bread ($p = 0.012$; $n = 70$), wheat ($p = 0.027$; $n = 67$), beef ($p = 0.040$; $n = 57$), mutton ($p = 0.034$; $n = 64$), fuel ($p = 0.017$; $n = 67$), and rent ($p = 0.067$; $n = 63$) correlate with relatively low fertility in 1920 in cases in which local fertility was not strongly dampened in 1919. There are also interaction effects in the expected direction in the cases of potatoes ($p = 0.086$; $n = 70$) and green beans ($p = 0.021$; $n = 67$). At high levels of increase in these prices, there is relatively little association between relative fertility in 1919 and 1920—as indicated by the narrowness of the difference between the predicted levels of relative 1920 fertility across the three terciles of relative 1919 fertility at the right side of the subplots in Figure 5. In other words, communities with high inflation for these goods in 1919 see relatively uniform relative fertility in 1920 without much respect to how strongly the pandemic seems to have dampened their fertility. At the low levels of increase in these prices, there is a strong positive association between relative fertility in 1919 and 1920. In other words, the 1920 baby boom in Spain was concentrated among communities that had relative stability or decreases in the prices of these goods in the year after the end of World War I. Additionally, non-interacting models indicate a negative association between price increases and 1920 fertility, controlling

for 1919 fertility, in the cases of the cost of electricity ($p = 0.052$; $n = 62$) and milk ($p = 0.065$; $n = 72$). These results suggest a strong importance of immediate post-war economic developments in shaping fertility in 1920, at least in the case of Spain.

The results shown in Figure 5 are somewhat cherrypicked. Table S1 shows the coefficient estimates, standard errors, and significance levels of the model terms for all 22 household prices, when allowing for and not allowing for interaction between 1919 fertility and price changes. However, we find them to be supportive of our overall argument. There are only two goods (ham and coffee) for which price increases are associated with higher fertility in 1920 at the $p < 0.10$ level of statistical significance—compared to the ten goods mentioned above for which the association is in the direction predicted by our hypothesis. It is also suggestive that the available data on household fundamentals like bread, milk, fuel, and rent fit our hypothesis.

There seems to be no economic dataset of comparable utility to that of Gómez-Tello et al. (2019) in any of the five other countries of interest, but as a second direct test of our hypothesis, we have looked at wage data from Norway. The wages of only six occupational groups were measured in both cities and rural locations by contemporary authorities. Additionally, it is not feasible to match wage data to the urban and rural parts of counties for which birth count data is available. However, averaging the relative changes in the available wage data between April 1918 and April 1919 at the county level yields further evidence supporting our hypothesis. The small sample sizes for each occupation ($n = 17–18$) make modelling an interaction effect impractical, but Table S2 shows that, when controlling for relative fertility in 1919, relative wage increases for two occupational groups are positively associated with relative fertility in 1920 at the $p < 0.05$ level. Wage increases for an additional two groups are significant at the $p < 0.10$ level.

These economic results are admittedly meagre, but we take them as support for our argument by exclusion that the boom was, in fact, caused by the end of World War I, given that it seems to have clearly not been caused by the end of the pandemic. In the future, these results should be reconsidered by quantitative and qualitative work that considers more closely the effect of short-term economic trends in this period on individuals' fertility behaviours.

Relevance to other pandemic research

Our key finding affirms the conclusion that different populations tend to be affected by pandemics in remarkably different ways. The 1918–20 influenza pandemic varied in its subnational severity (see e.g. Mamelund, 2003; Hayami 2015; Bambra et al. 2021), but populations and families also varied in their capacity to cope with the pandemic. The historical literature reports that some community institutions reacted fairly effectively to the pandemic (e.g. Keeling 2010; McLane 2013; León-Sanz 2014; Gallardo-Albarrán and de Zwart 2021), while others were paralysed to the detriment of those they served (e.g. Ellison 2003; Feldman 2014; Sobral et al. 2014; Rao and Greve 2018). The pandemic's effect on severe mental health outcomes like suicides seems to have been mostly null (see a review in Gaddy 2021; see also Chapelle 2022; cf. Strange 2022), but its adverse effects were wide-reaching in other domains. For example, recent econometric work concludes that the pandemic was causally associated with the economic betterment of the wealthy in Italy at the expense of the poor (Galletta and Giommoni 2022).

On the specific front of post-pandemic fertility, Wagner et al. (2020) find that cities in the United States that implemented longer periods of non-pharmaceutical intervention against the pandemic saw their fertility rebound to pre-pandemic levels

the fastest. Boberg-Fazlic et al. (2021) also find that conceptions in the two years after the peak of the pandemic in Sweden were disproportionately driven by married women and women with high social status surnames (see Tables B5 and B6). Like Covid-19, the 1918–20 pandemic was not a great equaliser (Mamelund and Dimka 2021), and our findings show yet another way in which that was true. The pandemic did not result in a bust-boom cycle that meant that its effects on fertility were effectively “egalitarian”.

Our findings are also relevant to the literature on the effect of in-utero exposure to the 1918–20 pandemic. It has been reported that the 1919 birth cohort in several countries was disadvantaged in terms of health or economic status relative to surrounding cohorts (e.g. Almond 2006; cf. Cohen et al. 2010). This has been interpreted as evidence for the pandemic influenza strain having adverse in-utero effects. However, Beach et al. (2022) have reported that this finding in the United States may be entirely attributable to the 1919 birth cohort having been born into disproportionately disadvantaged households. Floris et al. (in press) find small-scale but similar results in Bern, Switzerland. Similarly, Smith-Greenway et al. (2022) find that the disruptive effect of death can increase bystanders’ risk of unintended pregnancy far more than that of intended pregnancy, albeit in a contemporary context.

We additionally suggest that the 1920 cohort—which is used as part of the baseline comparison to 1919 in most studies considering the in-utero question—may have been born into relatively advantaged households. The findings of Boberg-Fazlic et al. (2021) about the social correlates of fertility in Sweden in 1920 make this point clearly. In our own work, we find that children born in 1920 in five out of the six national populations were disproportionately born into communities that had relatively high fertility in 1919 and that, therefore, can be assumed to have been socially affected only mildly by the pandemic. As discussed in our section on post-pandemic marriage

patterns, it also seems more likely than not that children born in neutral Europe in 1920 were born to parents who were married for longer and may have had more experience raising children than usual. Understanding which families were and which families were not having children during and in the wake of the pandemic may contribute to the debate around the impacts of in-utero exposure—and especially which baselines are appropriate for estimating the effects of exposure.

Conclusion

We offer the novel hypothesis that subpopulations whose fertility was hit hardest by the 1918–20 influenza pandemic saw below-average fertility in both 1919 and 1920. Subnational and cohort specific data from a total of six countries support this hypothesis. Our results suggest that the baby boom that occurred in much of neutral Europe in 1920 was caused by the end of World War I and especially its economic impacts, not by the waning of the pandemic. With the work of Mamelund (2004) cast in doubt, we know of no clear evidence, even at the ecological level, that individuals who postponed fertility during the main wave of the pandemic recuperated that fertility to a significant extent in 1920.

These findings are novel, and we hope they will prompt deeper research into the mortality-fertility nexus of the 1918–20 pandemic. As such, we would like to suggest several areas of research that might follow up on our work. To our knowledge, national CSFRs are not available for the period of interest in Spain, Switzerland, and the Netherlands, but subnational cohort-specific analysis in those countries may support or challenge our findings. Analyses of microdata may also contradict our conclusion, especially due to the possibility that some ecological fallacy lurks in our analysis. We discuss two mechanisms that have been suggested to us as alternate

explanations for our main results, but we may have left other possible mechanisms unexplored. Microdata data that would enable individual-level mechanisms to be studied more closely might be available for Iceland, where the families afflicted by pandemic mortality have already been mapped (see Gottfredsson et al. 2008). Additionally, there may have been a 1920 baby boom in some subpopulations of non-European countries, even though one does not appear at the national level; this would be like the case of Spain. It is also crucially important to study how fertility rebounded after the pandemic in indigenous and colonised populations, whose mortality from the pandemic was globally most severe by far (see e.g. Ohadike 1991; Adams 1997; Mamelund et al., 2013; Brady and Bahr 2014; Gara 2019; Rice 2019). In doing so, research can build on the work of Underwood (1984) and Herring (1993).

The mechanisms by which the pandemic affected individuals' fertility behaviours must also be better quantified and disentangled. The fact that, among the countries of neutral WEurope, Spain had the highest 1918 excess mortality (Echeverri 2003; Ansart et al. 2009) but the lowest relative fertility in 1920 (Figure 1) may suggest an importance of the direct effects of mortality on fertility, or this may be a coincidence. Our cohort analysis and the geographic analysis of Mamelund (2004) tentatively suggest that psychological or cultural reactions to the pandemic may have been key to the post-pandemic dampening of fertility. However, it is, therefore, puzzling that the dampening seems to have been strongest at older reproductive ages. One would expect younger groups to be more affected by any uncertainty or ideational change that the pandemic caused. It does not seem plausible that this effect is only due to a reduction in fecundity due to maternal ageing between 1919 and 1920.

To better disentangle how the pandemic impacted fertility in general, it will be necessary to understand the ways in which pandemic mortality, morbidity, and social

disruption affected different subpopulations' ability to either achieve or prevent childbearing. Doing so thoroughly will require interdisciplinary efforts and an integrated biosocial framework. All pandemic experiences are biosocial (see e.g. Fuentes 2020; Dimka et al. 2022), but a pandemic's effects on kinship and fertility are especially so. The 1918–20 influenza pandemic's effects on fertility are much less well understood than its death toll. Studying the former, especially through a biosocial lens, may yield key insights into both historical and modern populations' experiences of pandemics.

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Data availability statement

Our data and an R Markdown replication file for our analysis have been deposited in an Open Science Framework repository: https://osf.io/24d3e/?view_only=bd4acab9f0104f1e8a17395ddab00ba5

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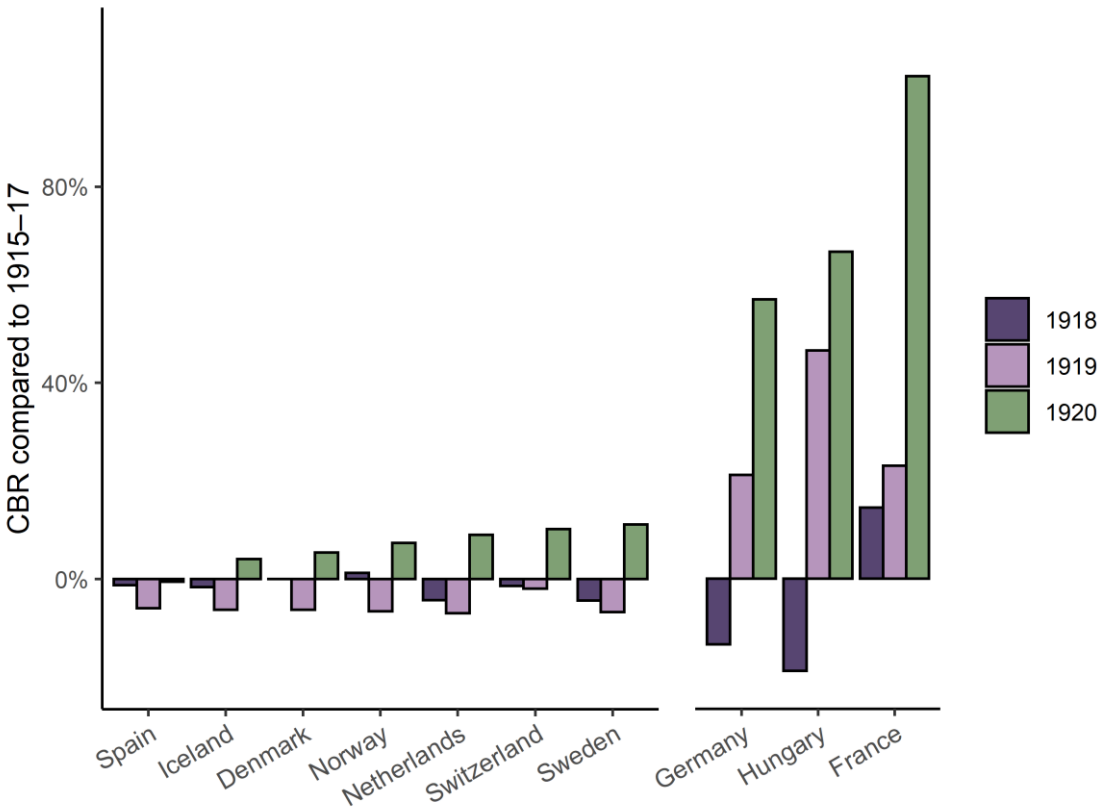

Figure 1. The relative crude birth rate (CBR) in 1918, 1919, and 1920 in seven neutral European countries (Spain, Iceland, Denmark, Norway, the Netherlands, Switzerland, and Sweden) and three belligerent European countries (Germany, Hungary, and France), using 1915–17 as a baseline.

Note: The data utilised come from the Human Mortality Database and Jónsson & Magnússon (1997) for Iceland; Chesnais (1992) for the other countries.

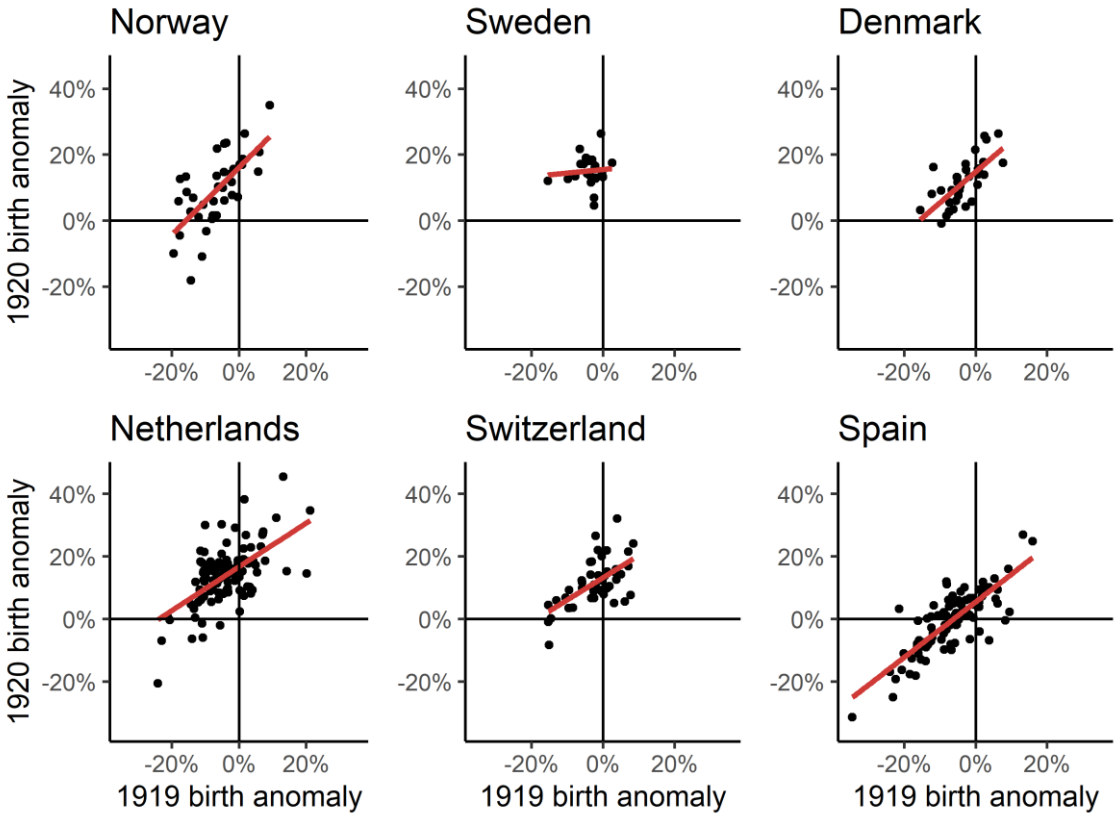

Figure 2. The correlation of the relative subnational birth counts in 1919 and 1920 in Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain, using 1916–18 as a baseline.

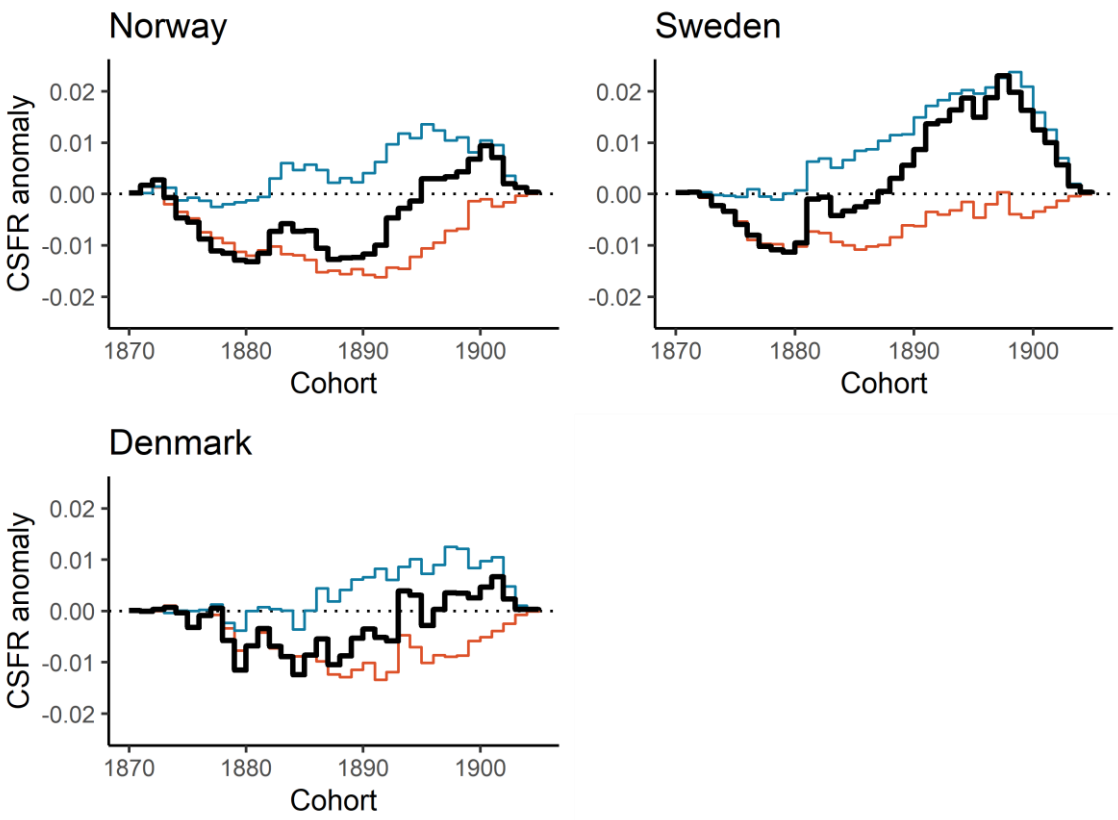

Figure 3. The cohort-specific fertility rate (CSFR) anomaly per woman in 1919 (red line) and 1920 (blue line) in Norway, Sweden, and Denmark, using 1916–18 as a baseline. The net anomaly over 1919–20 is also shown (thick black line).

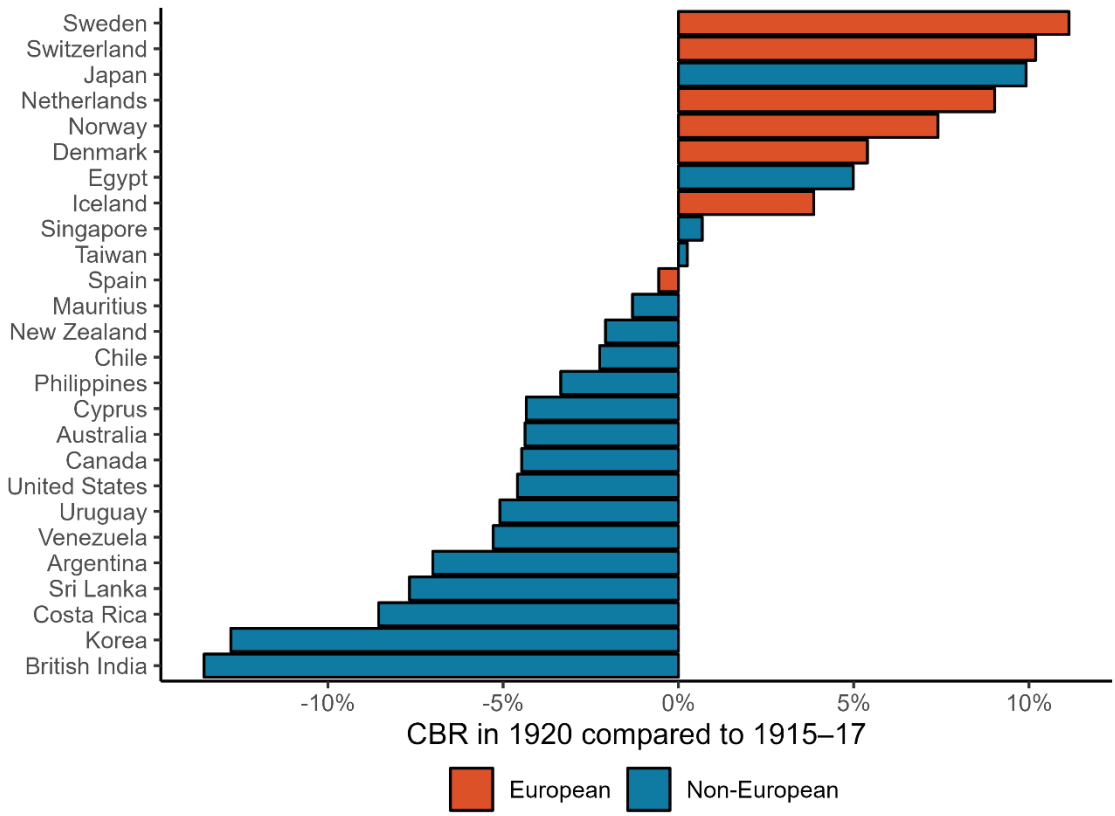

Figure 4. The relative crude birth rate (CBR) in 1920 in a range of non-European and neutral European countries, using 1915–17 as a baseline.

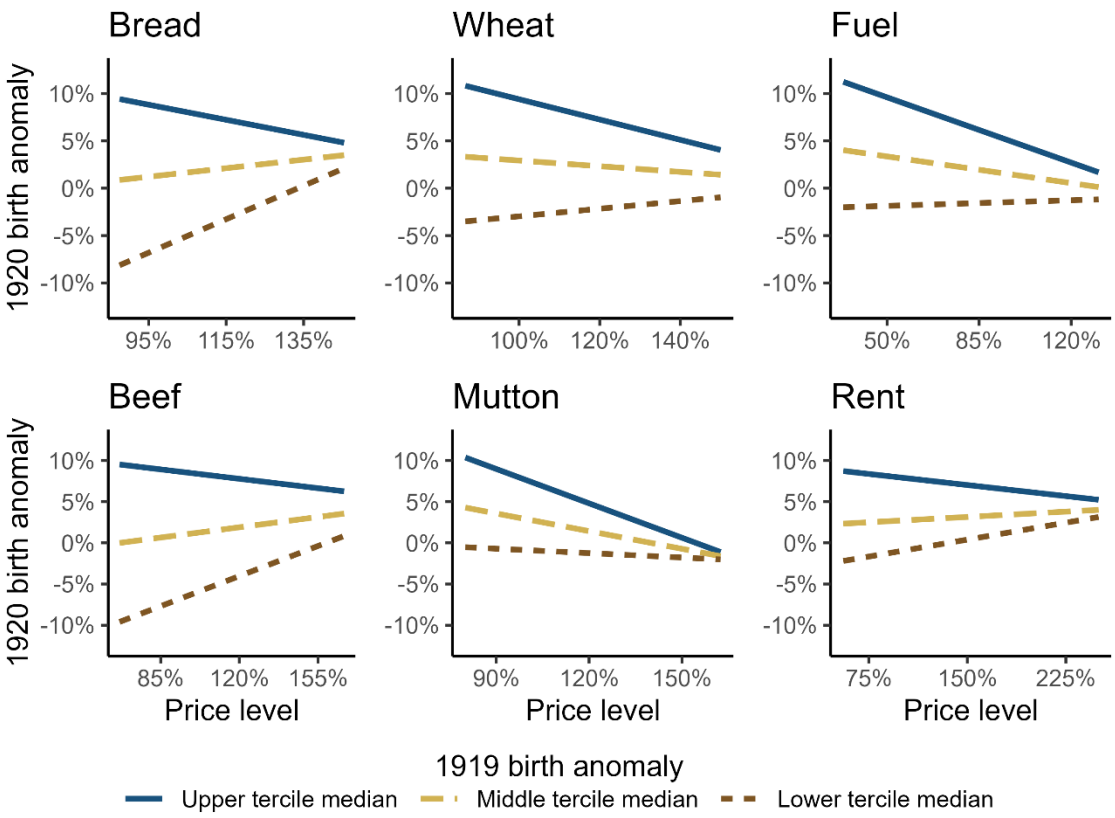

Figure 5. A set of interaction plots showing the linearly predicted value of the local 1920 relative birth anomaly in Spain (using 1916–18 as a baseline) based on the change in the price of bread, wheat, beef, mutton, fuel, and rent from winter 1918–19 and winter 1919–20 and different levels of 1919 relative birth anomaly.

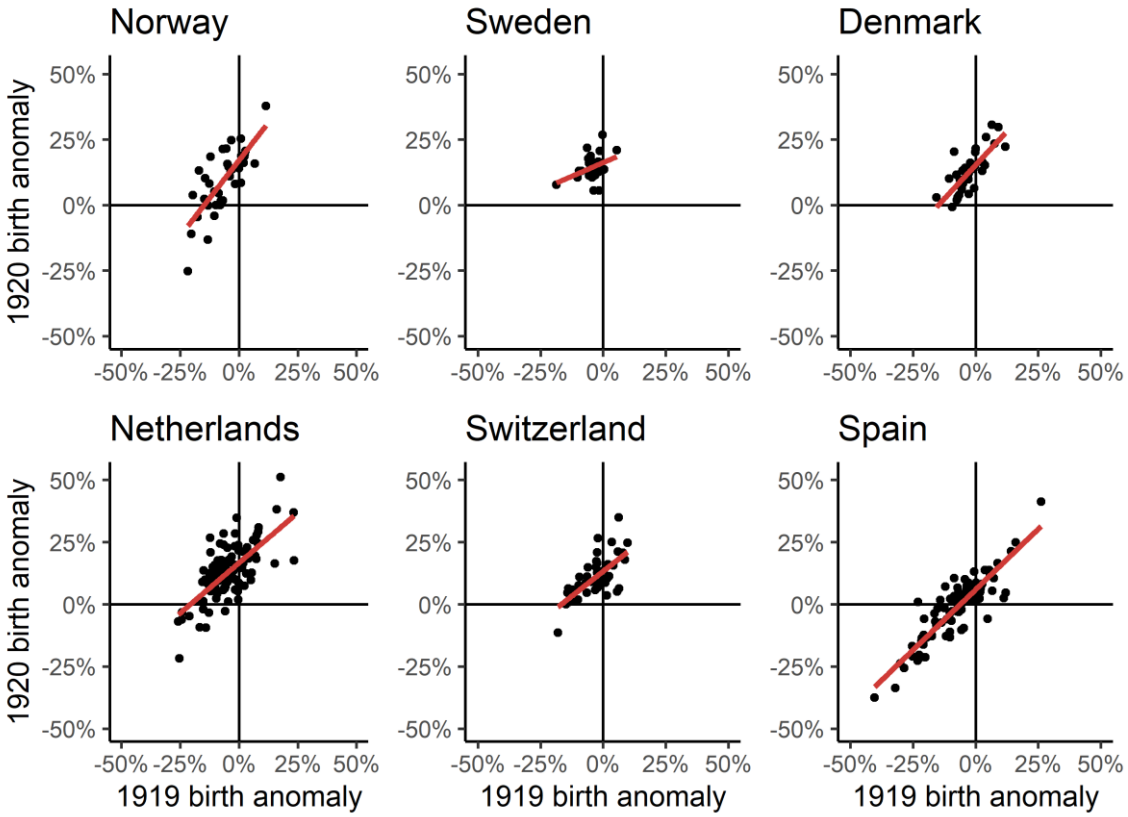

Figure S1. The correlation of the relative subnational birth counts in 1919 and 1920 in Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain, using 1916–17 as a baseline.

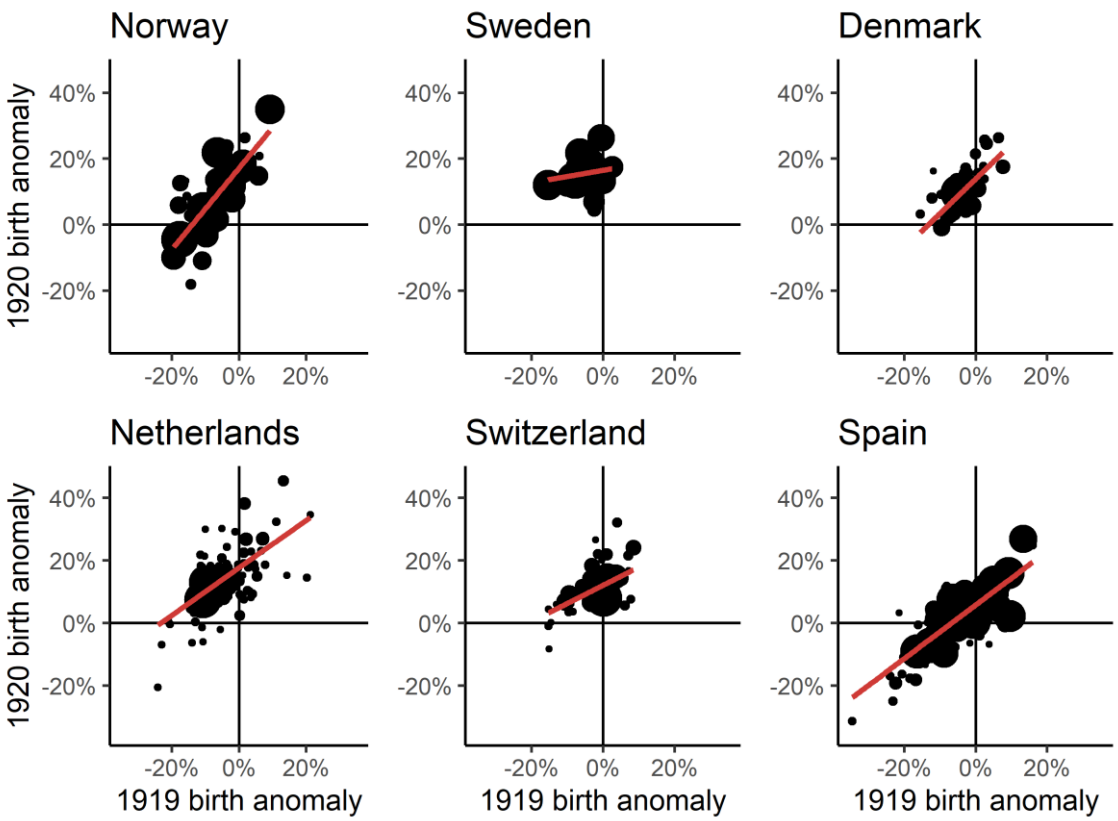

Figure S2. The correlation of the relative subnational birth counts in 1919 and 1920 in Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain, using 1916–18 as a baseline and the absolute size of the local birth count baseline as a regression weight.

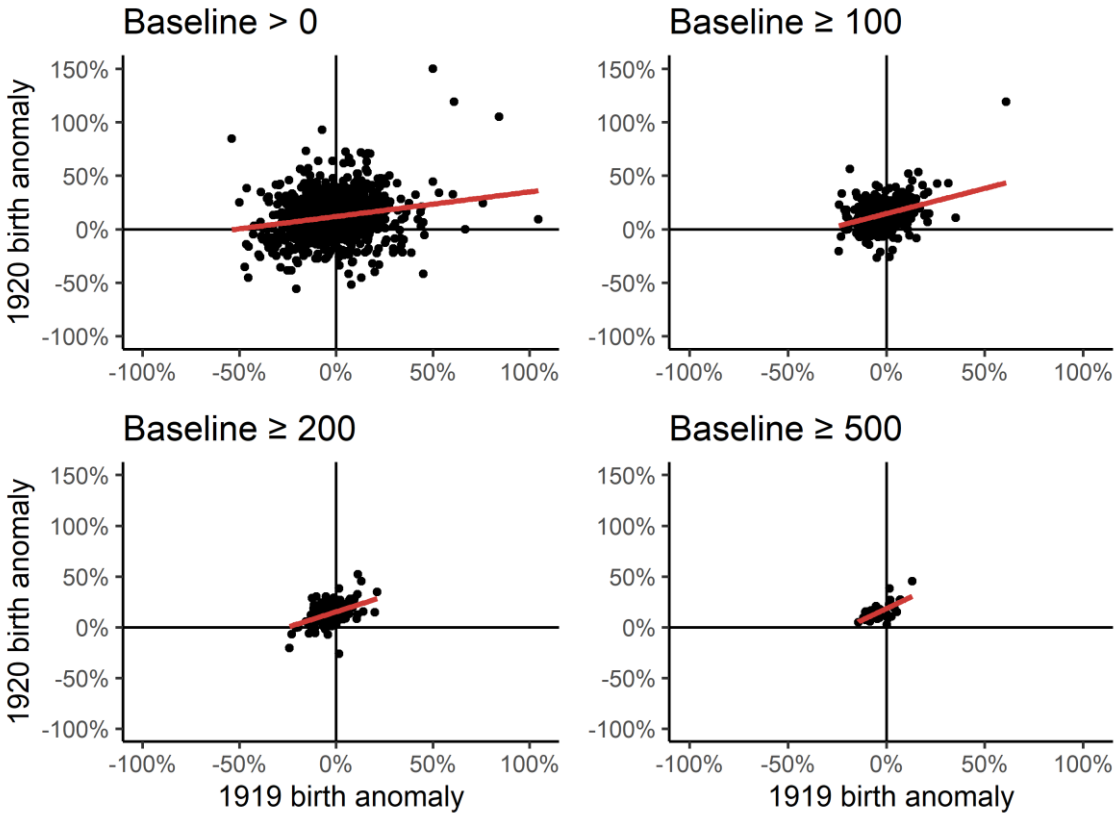

Figure S3. The correlation of the relative subnational birth counts in 1919 and 1920 in the municipalities of Netherlands, using 1916–18 as a baseline and different annual birth count baselines (> 0, ≥ 100, ≥ 200, and ≥ 500) as minimum size inclusion criteria for the municipalities.

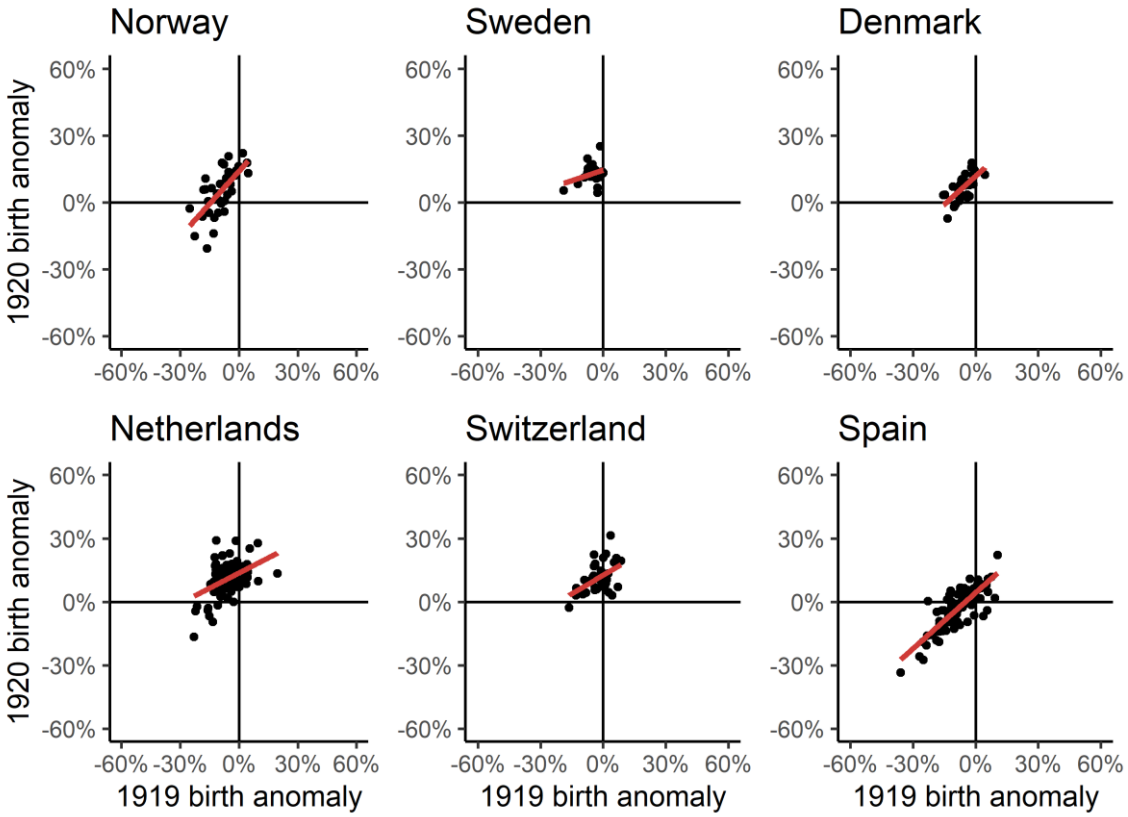

Figure S4. The correlation of the relative subnational birth rates in 1919 and 1920 in Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain, using 1916–17 as a baseline.

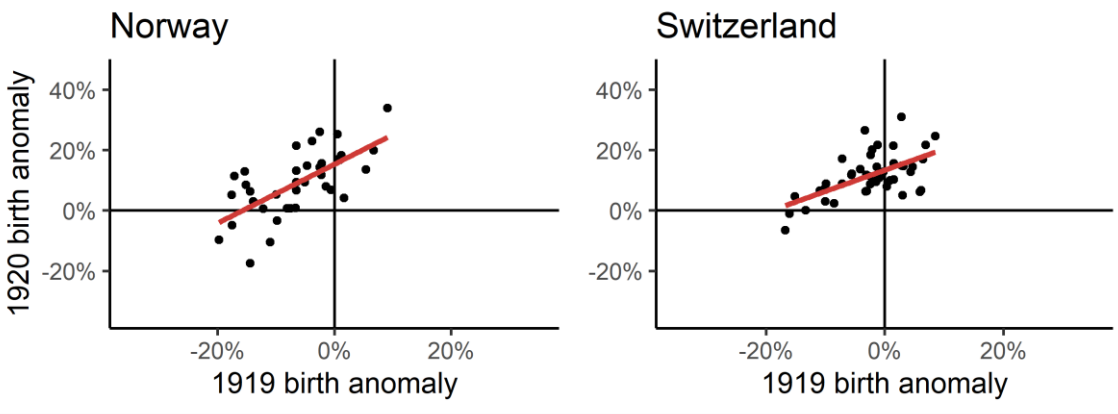

Figure S5. The correlation of the relative subnational birth counts in 1919 and 1920 in Norway, Sweden, Denmark, the Netherlands, Switzerland, and Spain, using 1916–18 as a baseline and including stillbirths.

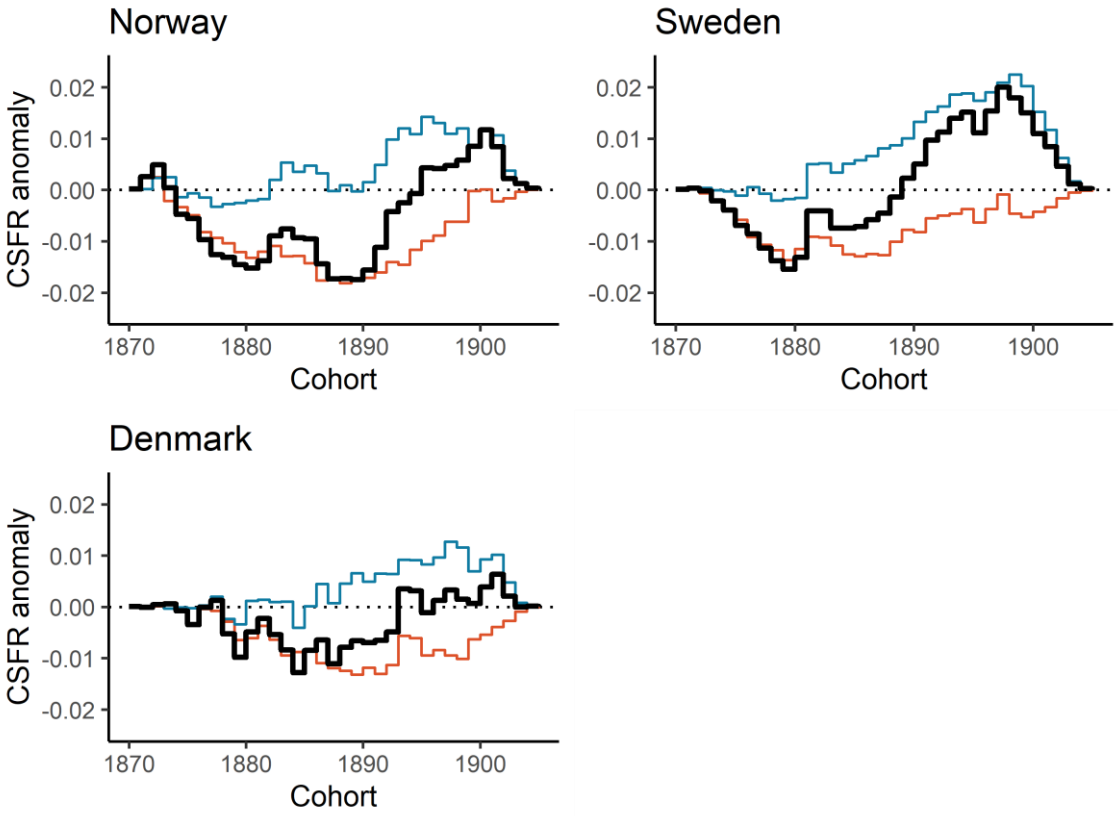

Figure S6. The cohort-specific fertility rate (CSFR) anomaly per woman in 1919 (red line) and 1920 (blue line) in Norway, Sweden, and Denmark, using 1916–17 as a baseline. The net anomaly over 1919–20 is also shown (thick black line).

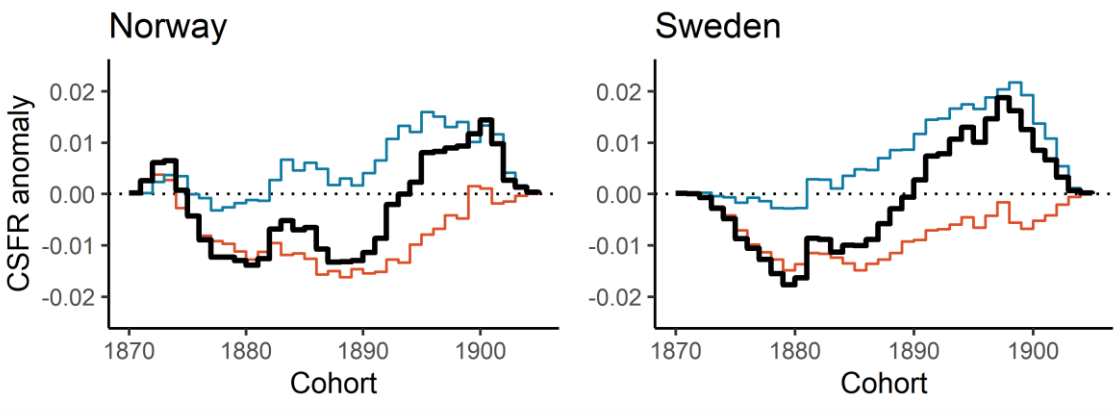

Figure S7. The cohort-specific fertility rate (CSFR) anomaly per woman in 1919 (red line) and 1920 (blue line) in Norway, Sweden, and Denmark, using 1915–17 as a baseline. The net anomaly over 1919–20 is also shown (thick black line).

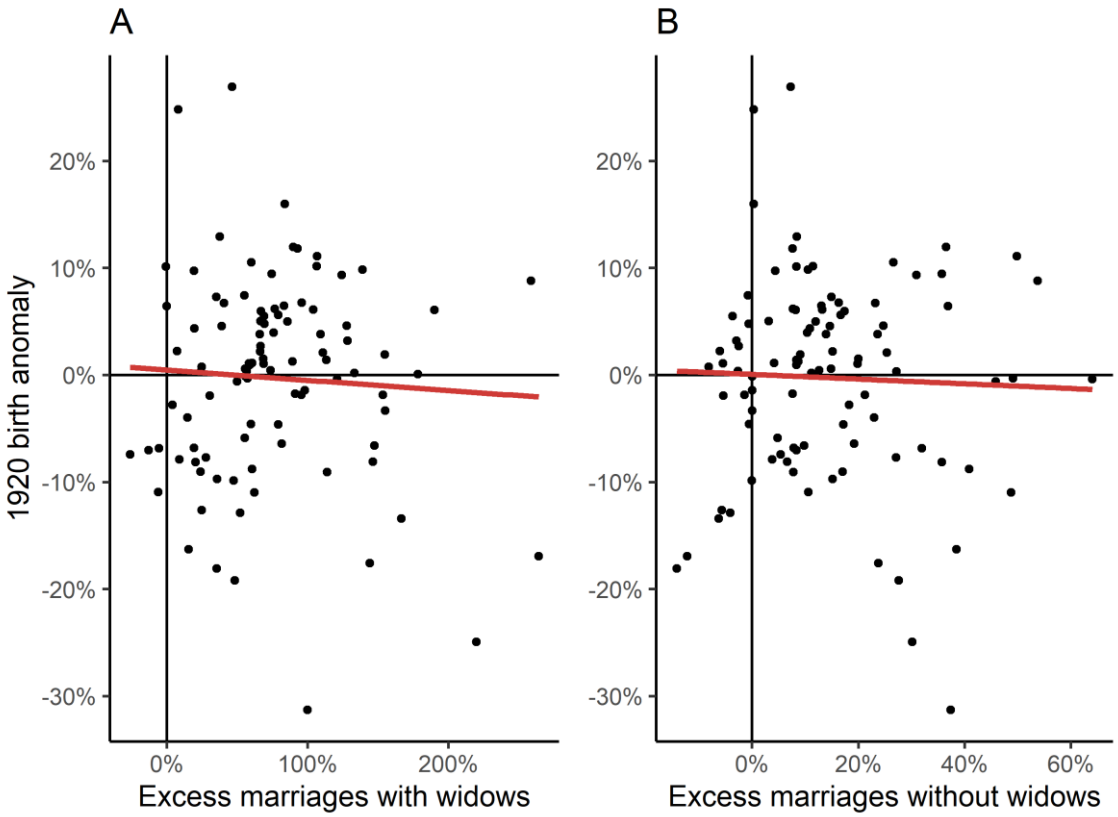

Figure S8. The correlation of the relative subnational anomaly in the number of marriages involving widows (A) and not involving widows (B) in 1919 (using 1917 as a baseline) against the relative subnational birth count anomalies in 1920 in Spain (using 1916–18 as a baseline).

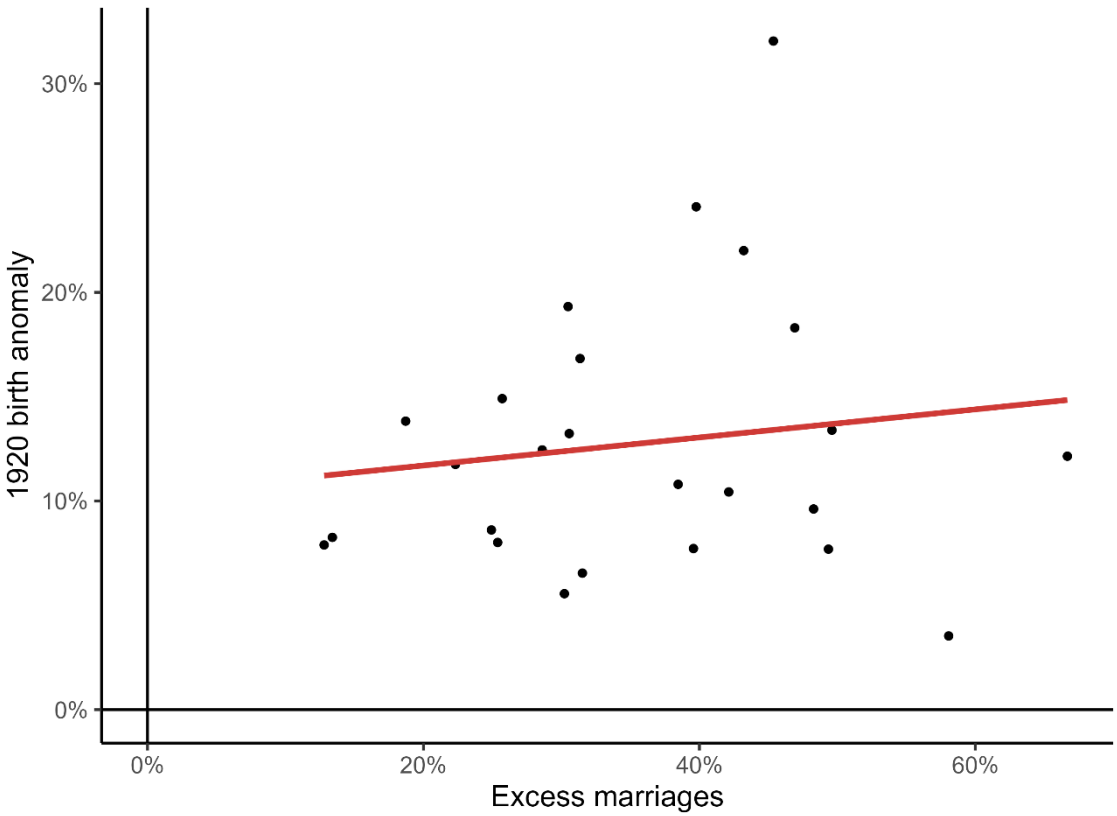

Figure S9. The correlation of the relative subnational anomaly in the number of marriages in 1919 (using 1917 as a baseline) against the relative subnational birth count anomalies in 1920 in Switzerland (using 1916–18 as a baseline).

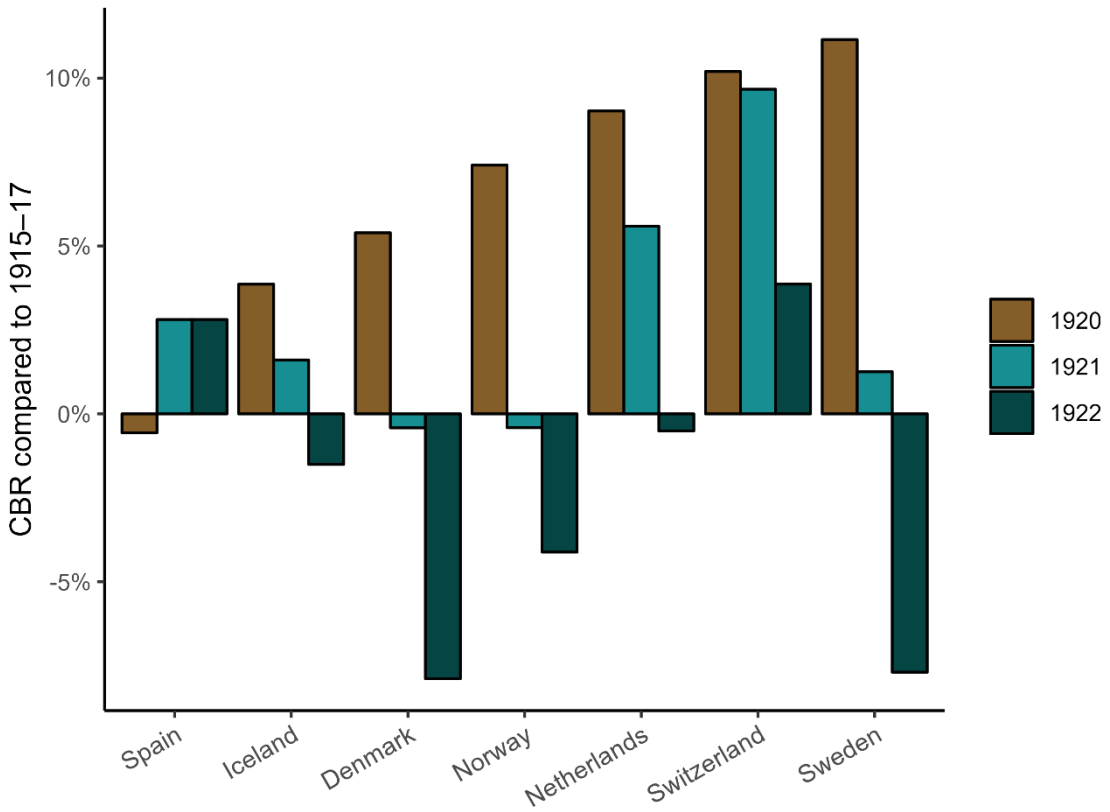

Figure S10. The relative crude birth rate (CBR) in 1920, 1921, and 1922 in seven neutral European countries (Spain, Iceland, Denmark, Norway, the Netherlands, Switzerland, and Sweden), using 1915–17 as a baseline.

Note: The data utilised come from the Human Mortality Database and Jónsson & Magnússon (1997) for Iceland; Chesnais (1992) for the other countries.

Table S1. A regression table testing the subnational association of relative price changes and the 1919 fertility anomaly with the 1920 fertility anomaly in Spain.

Note: Standard errors are in parentheses, and * indicates p < 0.10, ** indicates p < 0.05, and *** indicates p < 0.01

Dependent variable: 1920 fertility anomaly
Allowing interaction: No

Good	n	Intercept	1919 fertility anomaly	Price change
Bread	73	0.0243 (0.0639)	0.7493*** (0.0865)	0.0267 (0.0540)
Wheat	70	0.0807 (0.0541)	0.7697*** (0.0851)	–0.0199 (0.0445)
Beef	59	–0.0053 (0.0377)	0.8620*** (0.1024)	0.0633* (0.0346)
Mutton	66	0.0856** (0.0362)	0.6505*** (0.0945)	–0.0302 (0.0332)
Fresh fish	66	0.0656** (0.0255)	0.7723*** (0.0861)	–0.0081 (0.0220)
Salted cod	73	0.1085*** (0.0397)	0.7278*** (0.0864)	–0.0600 (0.0444)
Vegetables	60	0.0677*** (0.0148)	0.7824*** (0.0990)	–0.0066 (0.0097)
Potatoes	72	0.0645** (0.0315)	0.7560*** (0.0885)	–0.0074 (0.0266)
Chickpeas	72	0.0552* (0.0311)	0.7302*** (0.0867)	0.0007 (0.0231)
Rice	71	0.0442 (0.0593)	0.7738*** (0.0871)	0.0100 (0.0546)
Green beans	68	0.0057 (0.0339)	0.7062*** (0.0875)	0.0332 (0.0218)
Wine	73	0.0718* (0.0378)	0.7547*** (0.0874)	–0.0099 (0.0257)
Milk	73	0.1304*** (0.0404)	0.7444*** (0.0829)	–0.0646* (0.0344)
Fuel	68	0.0790*** (0.0225)	0.7228*** (0.0877)	–0.0329 (0.0306)
Electric light	64	0.0999*** (0.0256)	0.7363*** (0.0947)	–0.0383* (0.0194)
Charcoal	68	0.1168*** (0.0385)	0.7359*** (0.0876)	–0.0545 (0.0333)
Coffee	71	0.0033 (0.0463)	0.7333*** (0.0857)	0.0459 (0.0392)
Eggs	73	0.0337 (0.0365)	0.7258*** (0.0838)	0.0200 (0.0313)
Sugar	73	0.0276 (0.0610)	0.7922*** (0.0831)	0.0174 (0.0395)

Soap	71	–0.0242 (0.0413)	0.8608*** (0.0874)	0.0806* (0.0404)
Olive oil	73	0.0290 (0.0442)	0.7439*** (0.0878)	0.0209 (0.0340)
Rent for one room	64	0.0505** (0.0206)	0.6899*** (0.0882)	0.0067 (0.0160)

Dependent variable: 1920 fertility anomaly
Allowing interaction: Yes

Good	n	Intercept	1919 fertility anomaly	Price change	1919 fertility anomaly x Price change
Bread	73	0.1068 (0.0692)	2.7693*** (0.7832)	–0.0430 (0.0584)	–1.7756** (0.6846)
Wheat	70	0.1537** (0.0616)	2.1233*** (0.6044)	–0.0814 (0.0510)	–1.1544** (0.5106)
Beef	59	0.0771 (0.0535)	1.9594*** (0.5305)	–0.0133 (0.0494)	–0.9499** (0.4510)
Mutton	66	0.1768*** (0.0549)	1.8211*** (0.5488)	–0.1167** (0.0514)	–1.072** (0.4955)
Fresh fish	66	0.0433 (0.0319)	0.4289 (0.3081)	0.0122 (0.0280)	0.3080 (0.2654)
Salted cod	73	0.0986 (0.0730)	0.6206 (0.6683)	–0.0489 (0.0819)	0.1151 (0.7110)
Vegetables	60	0.0729*** (0.0160)	0.9351*** (0.2047)	–0.0105 (0.0107)	–0.1209 (0.1417)
Potatoes	72	0.0967*** (0.0361)	1.3698*** (0.3630)	–0.0353 (0.0307)	–0.5008* (0.2875)
Chickpeas	72	0.0706* (0.0385)	1.0062** (0.4126)	–0.0109 (0.0287)	–0.2081 (0.304)
Rice	71	0.1105 (0.0816)	1.6895** (0.7811)	–0.0513 (0.0753)	–0.8241 (0.6987)
Green beans	68	0.0536 (0.0384)	1.8217*** (0.4768)	0.0014 (0.0250)	–0.7450** (0.3134)
Wine	73	0.0373 (0.0568)	0.3058 (0.5574)	0.0136 (0.0387)	0.2910 (0.3568)
Milk	73	0.1181** (0.0480)	0.5251 (0.4604)	–0.0538 (0.0411)	0.1921 (0.3966)
Fuel	68	0.1133*** (0.0258)	1.4302*** (0.3001)	–0.0781** (0.0347)	–0.9097** (0.3703)
Electric light	64	0.1051*** (0.0272)	0.9050*** (0.2941)	–0.0424** (0.0206)	–0.1469 (0.2423)

Charcoal	68	0.0883 (0.0563)	0.3123 (0.6148)	–0.0293 (0.0492)	0.3750 (0.5388)
Coffee	71	–0.0435 (0.0578)	–0.0237 (0.5716)	0.0865* (0.0493)	0.6637 (0.4956)
Eggs	73	0.0314 (0.0411)	0.6693 (0.4656)	0.0220 (0.0353)	0.0502 (0.4070)
Sugar	73	0.0313 (0.0772)	0.8488 (0.7200)	0.015 (0.0497)	–0.0357 (0.4500)
Soap	71	–0.0664 (0.0487)	–0.1709 (0.6556)	0.1225** (0.0479)	0.9931 (0.6255)
Olive oil	73	0.0700 (0.0536)	1.4097*** (0.5071)	–0.0105 (0.0412)	–0.5368 (0.4027)
Rent for one room	64	0.0688*** (0.0224)	1.1918*** (0.2825)	–0.0084 (0.0176)	–0.4024* (0.2157)

Table S2. A regression table testing the subnational association of relative wage changes and the 1919 fertility anomaly with the 1920 fertility anomaly in Norway, using 1916–18 as a baseline.

Note: * indicates p < 0.10, ** indicates p < 0.05, and *** indicates p < 0.01

Dependent variable: 1920 fertility anomaly

Occupation	n	Intercept	1919 fertility anomaly	Wage change
Carpenter (Tømmermenn og bygningssnekkere)	18	–0.006 (0.097)	1.340*** (0.213)	0.141* (0.073)
Mason (Murere)	17	0.108 (0.075)	1.400*** (0.246)	0.052 (0.053)
Stone and earth workers (Cementstøpere o. a. anleggsarbeidere, sten- og jordarbeidere)	18	–0.052 (0.103)	1.331*** (0.205)	0.170** (0.074)
Servant aged 17-19 (Tjenestepiker)	18	0.057 (0.112)	1.346*** (0.229)	0.093 (0.083)
Servant aged 20-24 (Tjenestepiker)	18	–0.037 (0.102)	1.292*** (0.209)	0.148** (0.069)
Servant aged 25+ (Tjenestepiker)	18	–0.029 (0.104)	1.203*** (0.220)	0.138* (0.068)